

Lymphoid Tissue Histopathology - PRE-SAL Factoids

Week 7 SAL • no dedicated Word companion • update after class

Organ first, pathology second: identify architecture -> identify the normal compartment -> ask what is expanded, distorted, or effaced.

Organ fingerprints

Organ	Instant recognition	Main zone/function
Lymph node	Capsule + subcapsular sinus + cortex/paracortex/medulla	Filters lymph
Palatine tonsil	Squamous epithelium + deep crypts + follicles	Mucosal antigen sampling
Peyer's patch	Ileum + aggregated follicles + M-cell/dome region	GALT / IgA response
Appendix	Narrow lumen + colonic crypts + no villi + circumferential follicles	GALT
Spleen	Red/white pulp + central arteriole	Filters blood
Thymus	Lobules + dark cortex + pale medulla + Hassall corpuscles	T-cell maturation

Lymph node zones

Cortex	B-cell follicles ; secondary follicle = pale germinal center + darker mantle
Paracortex	T cells + HEVs ; lymphocytes enter from blood here
Medullary cords	Plasma cells, lymphocytes, macrophages
Medullary sinuses	Macrophage-rich lymph channels
Hilum	Artery, vein, efferent lymphatic

Primary follicle Dark, fairly uniform resting B-cell nodule; no pale germinal center .	Secondary follicle Pale germinal center + mantle zone; active B-cell proliferation, affinity maturation, class switching.
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Traffic cross-link

Antigen/DCs: afferent lymphatic -> subcapsular sinus -> cortex/paracortex. Naive B/T cells: blood -> HEV in paracortex. Exit: efferent lymphatic at hilum.

B/T map: node cortex B / paracortex T • spleen follicle B / PALS T • thymus = T-lineage organ, no B-cell follicles.

Reactive Lymphadenopathy - Provisional

Verify exact reactive patterns against the in-class SAL specimens.

Pattern	What expands	Recognition clue
Follicular hyperplasia	B follicles / germinal centers	Many variable-size secondary follicles; preserved mantle zones; polarity + tingible-body macrophages
Paracortical hyperplasia	T-cell zone	Expanded paracortex; immunoblasts may appear; follicles can be compressed
Sinus histiocytosis	Subcapsular/medullary sinuses	Dilated sinuses filled with histiocytes/macrophages
Suppurative lymphadenitis	Neutrophilic inflammatory foci	Neutrophils + necrosis / microabscesses
Granulomatous lymphadenitis	Macrophage aggregates	Epithelioid histiocytes +/- giant cells; necrosis depends on cause

Reactive vs neoplastic first pass

Reactive	Neoplastic concern
Overall architecture usually preserved	Architecture often effaced/replaced
Polymorphic cell population	Monotonous/repetitive population
Follicles vary in size/shape	Uniform back-to-back/repetitive follicles or sheets
Germinal-center polarity + tingible-body macrophages can be obvious	Loss of normal compartment relationships; clonal phenotype supports neoplasia

If you see X -> think Y

Deep squamous crypts	Palatine tonsil	Central arteriole + red pulp	Spleen
Ileal aggregated follicles	Peyer's patch	Hassall corpuscle	Thymus
Subcapsular sinus + cortex/paracortex/medulla	Lymph node	Big pale germinal center with tingible bodies	Reactive follicle can be normal response

Do not diagnose lymphoma from “big node” or “big germinal center.” Judge the whole architecture, compartment pattern, cellular diversity, and clonality.

No dedicated lymphoid-tissue summary exists. Normal histology comes from Week 7 lymphoid material; reactive patterns are a provisional pre-SAL framework until instructor slides are released.